8 Lecture 8 (January 31, 2025, est.): Tensor functors and moderate growth

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The parity element

We first make the condition on the element z from Lecture 7 explicit. Put $A = \mathcal{O}(G)$ and view $z \in G(k)$ as an even algebra homomorphism $z : A \to k$. The pullback of conjugation by z is

$$a \longmapsto \sum z(a_{(1)})\, a_{(2)}\, z^{-1}(a_{(3)}).$$

Here $z^{-1} = z \circ S$, and $z^2 = 1$ makes $z^{-1} = z$. The parity condition requires this expression to equal $a_0 - a_1$ for $a = a_0 + a_1$, $a_i \in A_i$. It does not require z to be central in the whole supergroup.

Monoidal and symmetric tensor functors

8.1 Definition. A *strong monoidal functor* $(F, F_0, F_2) : \mathcal{C} \to \mathcal{D}$ consists of a functor, an isomorphism $F_0 : \mathbb{1}_{\mathcal{D}} \to F(\mathbb{1}_{\mathcal{C}})$, and natural isomorphisms

$$F_{2,X,Y} : F(X) \otimes F(Y) \longrightarrow F(X \otimes Y).$$

They satisfy the associativity identity

$$F(a_{X,Y,Z})F_{2,X\otimes Y,Z}(F_{2,X,Y} \otimes \mathrm{id}) = F_{2,X,Y\otimes Z}(\mathrm{id} \otimes F_{2,Y,Z}) \circ a_{F(X),F(Y),F(Z)},$$

and the two unit identities

$$F(l_X)F_{2,\mathbb{1},X}(F_0 \otimes \mathrm{id}) = l_{F(X)},$$
$$F(r_X)F_{2,X,\mathbb{1}}(\mathrm{id} \otimes F_0) = r_{F(X)}.$$

For braided categories, F is *braided* if

$$F(c^{\mathcal{C}}_{X,Y})F_{2,X,Y} = F_{2,Y,X}c^{\mathcal{D}}_{F(X),F(Y)}.$$

If both braidings are symmetric, we call it a symmetric monoidal functor.

In these notes, a *symmetric tensor functor* between symmetric tensor categories means a k-linear exact strong symmetric monoidal functor. Such a functor is faithful.

Proof of faithfulness. If $X \neq 0$, the evaluation $X^* \otimes X \to \mathbb{1}$ is nonzero by the triangle identity, and hence surjective because the unit is simple. Exactness and $F(\mathbb{1}) \simeq \mathbb{1}$ imply $F(X) \neq 0$. If f is a nonzero morphism, its image is nonzero. Exactness gives $\operatorname{im} F(f) \simeq F(\operatorname{im} f) \neq 0$, so $F(f) \neq 0$. ∎

8.2 Lemma. *A symmetric tensor functor preserves categorical traces and dimensions.*

Proof. Applying F to evaluation and coevaluation, and inserting F_0, F_2, makes $F(X^*)$ a dual of $F(X)$. The triangle identities follow by applying F to those for X. Symmetry compatibility identifies the image of the braiding in the trace formula with the braiding of $F(X)$ and $F(X^*)$. Thus

$$\operatorname{Tr}(F(f)) = F_0^{-1} F(\operatorname{Tr}(f)) F_0.$$

On the scalar endomorphisms of the unit this is the same scalar because F is k-linear. Set $f = \mathrm{id}_X$ for the dimension statement. ∎

For $\operatorname{char} k \neq 2$, there is no symmetric tensor functor from sVec to $\operatorname{Rep}_k(G)$ for an ordinary affine group scheme G. Indeed, the odd line Π has tensor square $\mathbb{1}$ and categorical dimension -1. Its image would have one-dimensional underlying space, hence dimension 1, contradicting Lemma 8.2. The tensor-square condition is essential here: in positive characteristic an arbitrary vector space can have dimension congruent to -1 in k.

Moderate growth

Write $\ell(X)$ for the composition length of an object in a finite-length abelian category.

8.3 Definition. An object X has *moderate growth* if there is an integer $d_X \geq 1$ such that

$$\ell(X^{\otimes n}) \leq d_X^n \qquad (n \geq 0).$$

A tensor category has moderate growth if every object does.

8.4 Lemma. *For an exact faithful functor $F : \mathcal{C} \to \mathcal{D}$ between finite-length abelian categories, $\ell(X) \leq \ell(F(X))$.*

Proof. Apply F to a composition series of X. Exactness preserves its successive quotients, and faithfulness makes their images nonzero. Each image has length at least one; additivity of length gives the inequality. ∎

Consequently, moderate growth pulls back along tensor functors. In $\operatorname{Rep}(G, z)$ the ordinary total vector-space dimension, rather than superdimension, gives

$$\ell(X^{\otimes n}) \leq (\dim_k X_0 + \dim_k X_1)^n.$$

Categories with finitely many simple isomorphism classes also have moderate growth. To see this, choose representatives $S_1, \dots, S_r$. For fixed X put

$$d_X = \max\{1, \ell(X \otimes S_1), \dots, \ell(X \otimes S_r)\}.$$

Tensor exactness and a composition series of Y give $\ell(X \otimes Y) \leq d_X \ell(Y)$. Starting with $Y = \mathbb{1}$ proves the required bound by induction. In a finite tensor category one can alternatively use the Frobenius–Perron dimension: its positivity on simples gives $\ell(X^{\otimes n}) \leq \operatorname{FPdim}(X)^n$.

8.5 Theorem (Deligne)**.** *Let k be algebraically closed of characteristic zero and let $\mathcal{C}$ be a symmetric tensor category. The following are equivalent:*

1. *$\mathcal{C}$ has moderate growth;*
2. *there is a symmetric tensor functor $\mathcal{C} \to \mathrm{sVec}_k$;*
3. *$\mathcal{C} \simeq \mathrm{Rep}(G, z)$ as a symmetric tensor category for an affine supergroup G and an element z with $z^2 = 1$ and $\mathrm{Ad}(z)$ equal to parity.*

The implication from moderate growth to a super fiber functor is a substantial theorem, not proved here; see [EGNO15, Theorems 9.11.4 and 9.11.9]. The length estimate above proves the converse. The reconstruction behind the remaining equivalence motivates our next lectures, starting with ordinary fiber functors to vector spaces.

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